

BEYOND THE PLAN

3 Pillars, 1 Practice

A Frontline Safety Framework
for Temporary Traffic Management

Industry Position Paper

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*Prepared for industry consultation
with Main Roads Western Australia and traffic management organisations*

1. Executive Summary

Temporary traffic management in Western Australia is supported by comprehensive standards — the AGTTM, MRWA Code of Practice, and AS 1742.3 provide detailed guidance for planning, designing, and implementing traffic management at work zones. These standards represent significant expertise and are fit for their purpose: describing how a job should be set up and how traffic should be managed.

Yet workplace incidents in traffic management continue to occur on jobs where the paperwork was complete, the crew was accredited, and the signs were on the ute. The gap is not in the standards. The gap is between what the plan describes at the start of a shift and what the crew experiences during it. In safety science, this is known as the gap between "work as imagined" and "work as done" — and it is a gap that compliance-based systems cannot close.

This paper proposes a frontline safety framework — Beyond the Plan: 3 Pillars, 1 Practice — that names and structures what experienced traffic controllers already know: that the three things you verify before you start (competent crew, sound plan, right equipment) are necessary but not sufficient. The fourth element — dynamic awareness, the continuous practice of reading conditions and being willing to act on what you see — is what separates a safe crew from a crew that had all the right paperwork but still got hurt.

The framework draws on two complementary traditions in safety science. The three pillars are grounded in Safety I thinking — the established approach of identifying hazards, implementing controls, and verifying compliance. Dynamic awareness is grounded in Safety II thinking — the emerging paradigm that focuses on building adaptive capacity, studying how work succeeds, and treating workers as the solution rather than the variable to control. The framework does not choose between these traditions. It uses each where it is strongest: Safety I for the preconditions, Safety II for the practice that sustains them.

The framework is grounded in evidence from MRWA Banner Alerts, mapped against existing legislative and regulatory requirements, and designed for practical adoption without adding compliance burden. It does not replace current standards. It provides the connecting tissue that makes them work in real time, on the road, when conditions change.

Key proposition: The three pillars are preconditions — without them, the job should not proceed. The practice is continuous — without it, the preconditions degrade as conditions change. Communication is the thread that holds all four together. And the organisation is accountable for creating the conditions where all of this is possible.

2. The Problem: When the Paperwork Is Perfect

2.1 The Gap Between Plan and Reality

A Traffic Management Plan is a snapshot. It describes the work zone as it was designed — the sign positions, the speed zones, the traffic control points. It assumes the weather stays the same, the traffic behaves predictably, and the hazards are known in advance. These assumptions hold at the moment the plan is written. They do not necessarily hold at 2:30 PM on a Tuesday when the wind picks up, the sun shifts, and the traffic volume doubles because a nearby school just finished for the day.

This is not a failure of the planning process. Plans cannot predict every condition. The purpose of a plan is to establish a safe baseline. The problem arises when the baseline is treated as the complete picture — when the crew follows the plan right up to the moment someone gets hurt, and the investigation reveals that conditions had changed and nobody adjusted.

The safety science literature has a name for this gap. Erik Hollnagel, one of the founders of Resilience Engineering, distinguishes between "work as imagined" — what the procedure, the plan, and the risk assessment assume will happen — and "work as done" — what actually happens when real people do real work in real conditions. The gap between the two is not an anomaly. It is a permanent feature of complex work. The question is not whether the gap exists, but whether the system is designed to cope with it.

2.2 The Evidence: Banner Alert Case Studies

The MRWA Banner Alert system provides documented evidence of incidents that occurred on jobs where the standard preconditions were met. The following cases illustrate the gap:

Case 1: TC Slips on Soft Roadside Batter (EQ#57872)

A Traffic Controller exited their vehicle on the passenger side at 6:15 AM. Their feet slipped on a soft batter at the road shoulder. They grabbed the door and roll bar to right themselves, sustaining a shoulder injury that resulted in a Lost Time Injury.

Pre-start status: The crew was accredited. The TMP was in place. The signs were loaded and positioned. All three pillars were satisfied.

What the plan didn't account for: Low light at dawn combined with a soft roadside batter that looked stable but wasn't. This hazard only existed in that moment — it was not present when the site was assessed in daylight, and it would not be present after the ground dried. Dynamic awareness — scanning

the ground surface before stepping out, especially in low light — would have identified the hazard before it caused injury.

Case 2: Core Tester Working Within 1.2m of Live Traffic at 110 km/h (EQ#39231)

A lab core tester was conducting asphalt core sampling within 1.2 metres of live traffic travelling at 110 km/h on a Western Australian highway. The approved Traffic Guidance Scheme (TGS-3) required speed reduction signs to be in place before work could commence within 1.2m of the live lane. Instead, the traffic controllers provided only a shadow vehicle. When instructed by the Traffic Management Coordinator to remain off the road until the correct TGS was implemented, the core tester did not comply and continued working. The project engineer had departed site at 11:00 AM for the holiday break, leaving inadequate supervision. The core tester's SWMS was an outdated revision from the previous June, and they had not reviewed or signed onto the current version. The potential consequence was assessed as Major — a vehicle at 110 km/h striking a worker on foot would almost certainly be fatal.

Pre-start status: The crew held accreditations. A TMP existed with an approved TGS-3. Signs were on site. On paper, the three pillars were satisfied.

What the plan didn't account for: Time and productivity pressure — the core testing had to be completed within 24 hours of the asphalt being laid, on the last day of the rostered swing, with equipment faults already causing delays. The supervision gap — the project engineer had already left for the holiday break. The compliance gap — the traffic controllers substituted a shadow vehicle for the required speed reduction signs, and the core tester ignored the TMC's instruction to stand down. Every pillar was broken in practice while appearing intact on paper. Dynamic awareness would have required the TMC to halt work until TGS-3 was correctly implemented, and would have empowered the core tester to refuse to proceed without proper protection — but the system incentivised getting the job done over getting it done safely.

Case 3: Sun Glare Changes Mid-Job (EQ#57762)

A traffic management job started with clear visibility. As the afternoon progressed, the sun shifted to a position that caused direct glare into the Traffic Controller's line of sight. The TC could no longer see oncoming traffic clearly, creating a significant safety risk.

Pre-start status: The TMP was correct for the conditions at the start of the shift. Signs were positioned properly.

What the plan didn't account for: The sun's position at 3:30 PM. The plan was written for the setup time, not the entire shift. Dynamic awareness — recognising that visibility had degraded and triggering a reassessment — would have led to repositioning the TC, adding a spotter, or pausing until the glare passed.

Case 4: TC and Work Vehicle Within 1.2m of Live Traffic (EQ#39486)

A Traffic Controller and their work vehicle were found positioned within 1.2 metres of a live traffic lane during an active traffic management operation. The traffic control setup had not been verified correctly before work commenced. The Traffic Controllers were not fully aware of the hazards or how to implement effective controls as specified in the Safe Work Method Statement (SWMS). There was no evidence of ongoing training and support by the service provider for the Traffic Controllers on the crew. The setup also included an unapproved "add-on" to the traffic management plan that was not condemned by the Team Leader. This incident scored the maximum relevance score of 5 under the three pillars assessment — all five failure indicators were flagged: conditions changed mid-job, crew competency gaps, plan adequacy failures, equipment issues, and the paperwork-vs-reality gap.

Pre-start status: The crew held Traffic Controller accreditations. A traffic management plan was in place. A SWMS existed. On paper, all three pillars were satisfied.

What the plan didn't account for: The TCs didn't fully understand the hazards they were working within or the controls specified in the SWMS. The traffic control setup was never verified. An unapproved modification to the plan was made on the fly and the team leader allowed it. The service provider had no evidence of providing ongoing training or support to its TCs. This case illustrates the deepest form of the gap: the paperwork existed but was not understood by the people whose safety depended on it, the setup was not checked against the plan, and the culture allowed unapproved deviations. Dynamic awareness would have required the TC to question their position relative to live traffic, the team leader to verify the setup before work began, and any crew member to flag the unapproved add-on — but none of these happened because the system had trained people to follow orders, not to think critically about their own safety.

2.3 The Pattern

Across these incidents, a consistent pattern emerges:

- **Compliance:** The three preconditions were satisfied on paper
- **Change:** Conditions changed during the job, or a hazard existed that was not identified in the plan
- **Gap:** The crew did not adjust — either because they didn't notice the change, didn't feel authorised to act, or didn't have a process for reassessment

This pattern is not unique to Western Australia. It is consistent with safety research across high-risk industries, from construction to aviation to offshore operations. The pattern has a name in the safety literature: the gap between "work as imagined" (what the plan assumes) and "work as done" (what actually happens). The question is not whether this gap exists — it always does. The question is whether the crew has the awareness, the authority, and the process to bridge it.

3. The Intellectual Foundation: From Safety I to Safety II

This framework does not emerge from nowhere. It sits at the intersection of two traditions in safety science — one established, one emerging. Understanding both is essential because the 3 Pillars draw on the first tradition, while the Practice draws on the second. The framework's strength is in using each tradition where it is strongest.

3.1 Safety I: The Established Tradition

Safety I is the approach most safety professionals were trained in. It underpins the DuPont safety system, the MRWA Code of Practice, and virtually every compliance framework in Australian traffic management. Its core assumption is that safety is the absence of negative outcomes — no injuries, no incidents, no non-conformances. When something goes wrong, Safety I asks: what broke? What caused the failure? How do we prevent it from happening again?

The DuPont safety system, developed over more than 200 years since the company began manufacturing gunpowder in 1802, is the most influential expression of Safety I thinking in industrial safety. DuPont's Front Line Management course, widely adopted across high-risk industries, teaches supervisors to manage safety through behavioural observation, compliance auditing, and root-cause investigation. The Bradley Curve — DuPont's cultural maturity model describing the progression from Reactive through Dependent and Independent to Interdependent — remains the most widely recognised framework for assessing safety culture maturity.

Safety I has achieved remarkable results. Industries that adopted it — chemical manufacturing, oil and gas, aviation — saw dramatic reductions in workplace fatalities and serious injuries. In traffic management, Safety I gives us the prestart checklist, the TMP, the SWMS, and the audit regime. These are not trivial achievements. They represent a baseline of systematic safety practice that prevents many categories of harm.

3.2 The Limitations of Safety I in Traffic Management

Safety I was developed for controlled environments — chemical plants, factories, refineries — where the workplace is enclosed, the workforce is permanent, and conditions are stable. Temporary traffic management is fundamentally different in four ways:

- **Uncontrolled environment:** The environment changes constantly. DuPont plants are the same every day. A TC work zone is different every setup. Weather, traffic, light, road surface, and adjacent construction activity can all change within a single shift.

- **Public in the hazard zone:** The public drives through the workplace at 80 km/h. DuPont controls who enters the plant and how they behave. TCs have no control over the motorists who pass through their work zone, often at speed and sometimes under the influence, distracted, or confused by the layout.
- **Mobile and temporary work:** TCs set up, work, and tear down, sometimes multiple times a day. There is no "steady state" to optimise. Each setup is a new configuration with new risks.
- **Casualised workforce:** The workforce is often labour-hire, short-term, and high-turnover. DuPont has career employees who develop deep institutional knowledge over decades. Many TCs are on their first or second season, working alongside people they met this morning.

These differences matter because they mean that the gap between "work as imagined" and "work as done" is wider, more variable, and less predictable in traffic management than in the environments Safety I was designed for. A compliance system that works in a chemical plant — where conditions are stable and deviations are rare — struggles in a traffic management context where conditions change continuously and deviation from the plan is often the only way to stay safe.

3.3 Safety II: The Emerging Paradigm

Since approximately 2014, a new paradigm has emerged in safety science that directly addresses the limitations of Safety I. Three thinkers and their frameworks form the core of this shift:

Erik Hollnagel: Safety II and Resilience Engineering

Hollnagel's foundational question is the simplest and most profound: instead of asking "what went wrong?" why don't we ask "what goes right?" His insight is that safety is not the absence of negatives — it is the presence of capacities. A crew that successfully navigates a shift where the wind blows signs over, traffic ignores speed zones, and the sun blinds the TC is demonstrating safety. Not because nothing bad happened, but because they had the capacity to adapt and succeed.

	Safety I (Traditional)	Safety II (Hollnagel)
Core question	What went wrong?	What goes right?
Safety is defined as	Absence of negatives (no injuries)	Presence of adaptive capacity
Humans are	The variable to control	The reason things go right
Approach	Prevent things from going wrong	Ensure things go right more often
After an incident	Find what broke	Understand how people were trying to succeed

Hollnagel's Resilience Engineering framework identifies four capacities that enable systems to cope with surprise: the ability to respond to what happens, to monitor what is happening, to anticipate what may happen, and to learn from what has happened. These four capacities map directly to the components of

Dynamic Awareness: reading conditions (monitor), adjusting controls (respond), experience-based expectation (anticipate), and post-job review (learn).

Sidney Dekker: Safety Differently

Dekker builds on Hollnagel with three provocations that challenge the foundations of compliance-based safety. First, safety is not defined by the absence of negatives — it is defined by the presence of capacities. Second, safety is not a bureaucratic accountability — it is an ethical responsibility. Third, and most importantly, people are not the problem to control — they are the resource to harness.

Dekker's most significant practical contribution is the concept of Restorative Just Culture. When something goes wrong, compliance-based systems ask "who violated the rule?" Dekker asks "who has been hurt, and what do they need?" The shift from justice as retribution to justice as restoration is profound. In traffic management, it means moving from a culture where the crew leader who calls a stop gets asked "why aren't you following the plan?" to a culture where they get asked "what did you see that the plan didn't account for?"

Dekker's 2018 book "The Safety Anarchist" argues that more rules, more paperwork, and more bureaucracy can actually reduce safety by pushing risk underground and removing workers' ability to adapt. This is directly observable in TC operations: when the prestart becomes a form to complete rather than a conversation to have, it stops serving its safety purpose and becomes a compliance exercise that teaches crews that the paperwork matters more than the practice.

Todd Conklin: Human and Organisational Performance (HOP)

Conklin is the most practical of the three. His Five Principles of Human Performance provide a direct operational framework:

- **Error is normal:** Humans make mistakes, even well-trained, well-meaning humans. Stop designing systems that assume they won't.
- **Blame fixes nothing:** Punishing the person doesn't fix the system that allowed the error.
- **Systems drive behaviour:** People do what the system incentivises, not what the procedure says.
- **Learn from normal work:** Don't just investigate failures. Study how work actually happens day-to-day.
- **Failure is resurfacing normal variability:** Accidents come from normal variations that combine differently this time, not from broken parts.

Conklin's key practical tools are Learning Teams and Pre-Accident Investigations. Learning Teams replace root-cause investigations with facilitated conversations about how work actually gets done. Pre-Accident Investigations study normal work before failures occur. Both are directly applicable to TC crews: instead of waiting for an incident to investigate, sit down with the crew after a normal shift and ask "what did you have to do today to make this work?"

3.4 The Structural Contradiction in Australian Traffic Management

There is a contradiction at the heart of Australian traffic management safety that the framework is designed to address. Australia is a global leader in Safe System thinking for road users. The National Road Safety Strategy 2021-2030, Austroads frameworks, and MRWA's own road design standards all embrace the principle that humans are fallible and systems should be designed to tolerate error. This is Safety II thinking applied to the public.

But for the workers who set up and manage traffic control? The approach remains overwhelmingly Safety I: prescriptive TMPs, compliance audits, behavioural observation, root-cause investigation. No Australian traffic management company or road authority appears to have formally adopted Safety Differently, Safety II, or HOP as their primary safety framework for traffic workers.

This creates a structural contradiction: the road safety strategy says "humans make mistakes, design the system to cope," but the safety management of the people implementing that strategy says "comply with the plan, don't deviate, all injuries are preventable." The 3 Pillars, 1 Practice framework bridges this contradiction. The three pillars provide the Safety I baseline — verifiable, auditable, necessary. Dynamic awareness provides the Safety II practice — continuous, adaptive, behavioural. Together, they apply the same Safe System principle to workers that Australia already applies to road users.

4. The Framework: 3 Pillars, 1 Practice

The framework is built on a single observation: the things you verify before a job starts are different in kind from the thing you do while the job is happening. The three pillars are static — they are preconditions that, once verified, remain stable for the duration of the task. They are grounded in Safety I thinking — they are auditable, verifiable, and compliance-oriented. The practice is continuous — it begins when you arrive and does not stop until you pack down. It is grounded in Safety II thinking — it is behavioural, adaptive, and capacity-oriented.

Pillar 1: Competent Crew

The question: Do we have people who know what they're doing?

Competence is not the same as accreditation. A Traffic Controller with a valid BWMT card has met the minimum training requirement, but competence on a specific job requires more: familiarity with the road, experience with the type of setup, and the judgement that comes from having worked in conditions similar to those expected on the day.

The distinction matters because accreditation is binary (you have the card or you don't) but competence is contextual. A TC who is highly competent on metropolitan signalised intersections may not be competent on a rural highway at night. A TC who has installed hundreds of lane closures may not be competent on a mobile convoy operation. Competence is always relative to the task, the location, and the conditions.

What this pillar requires:

- All crew members hold current accreditations appropriate to their role
- At least one crew member has experience with the specific type of setup required
- The crew has been briefed on the site-specific risks, not just the generic hazards
- Any crew member new to this type of work is paired with an experienced hand

It's broken when: The crew is accredited but nobody has worked this road before, or a TC is assigned to a setup type they've never done without supervision.

Pillar 2: Sound Plan

The question: Do we know what we're doing and how?

A sound plan is one that is specific to the site, current for the conditions, and understood by the crew. It is not a generic TGS pulled from a folder. It is not a TMP that was written for a different time of day, a different traffic volume, or a different road configuration. And it is not a document that sits on the dashboard unread while the crew sets up from memory.

The MRWA Code of Practice and AGTTM provide the design standards for traffic management plans. This pillar is not about the quality of those standards — it is about whether the plan that reaches the crew is fit for the specific job they are about to do. A plan that was correct when it was written but does not account for current conditions is not a sound plan for today.

What this pillar requires:

- The TMP and TGS are site-specific, not generic
- The plan accounts for the time of day, expected traffic volumes, and known environmental conditions
- The crew has read and understood the plan, not just received it
- The plan includes provisions for conditions that may change during the shift (weather, light, traffic)

It's broken when: The TMP is generic and doesn't match site conditions, the TGS was written for a different setup, or the crew hasn't actually read the plan.

Pillar 3: Right Equipment

The question: Do we have what the plan requires?

Having the right equipment means more than loading the signs listed on the TGS schedule. It means the signs are serviceable (not faded, damaged, or wrong-sized), the supplementary equipment is available

(speed feedback signs, lighting towers, night wands), and there are sufficient reserves to adapt if conditions change during the shift.

The MRWA Code of Practice Section 6.16 mandates speed feedback signs at static worksites on Main Roads where workers on foot are not protected by barriers. Section 6.14 requires that all signs and devices are maintained in a serviceable condition throughout the work period. This pillar is about verifying compliance with these requirements before the job starts, not discovering their absence during it.

What this pillar requires:

- All signs and devices required by the TGS are loaded and in serviceable condition
- Supplementary equipment specified in the TMP is available (speed feedback signs, lighting, communication equipment)
- Equipment for contingencies is available (spare signs, additional delineation, wet weather gear)
- The crew has verified the equipment list against the plan, not just loaded from habit

It's broken when: Signs are loaded but the wrong size or faded beyond readability, the speed feedback sign that the TMP requires is back at the depot, or the crew loaded from memory without checking the schedule.

The Practice: Dynamic Awareness

"Dynamic Awareness is the practice of continuously reading the job while it's happening — and being willing to act on what you see."

— Beyond the Plan Framework

Dynamic awareness is not a checklist. It is not something you tick off at the pre-start brief. It is the only element of the framework that is active for the entire duration of the job, from the moment the first crew member steps out of the vehicle to the moment the last sign is loaded back onto the ute.

The three pillars are static — once verified, they remain stable. Dynamic awareness is continuous. It exists precisely because the three pillars, no matter how well they are satisfied at the start, will degrade as conditions change. The wind picks up. The sun shifts. The traffic volume changes. A crew member fatigues. A hazard that wasn't there yesterday appears. The plan doesn't cover it. That's where dynamic awareness becomes the safety net.

Dynamic awareness is a Safety II concept. Where Safety I asks "did you follow the procedure?" Dynamic Awareness asks "what did you see, and what did you do about it?" Where Safety I treats deviation from the plan as non-compliance, Dynamic Awareness treats it as adaptation — provided it is informed, proportionate, and communicated. The MRWA TMP risk register section 3.2.1.2 already authorises this: "Traffic Management Personnel may adjust or change the site where a significant safety hazard is identified." The framework gives this buried sentence prominence and process.

The Three Components of Dynamic Awareness

1. Reading Conditions

This is the observational foundation. Reading conditions means paying attention to what is actually happening, not what the forecast said this morning or what the TMP assumed. It includes monitoring weather changes (wind, rain, visibility), traffic behaviour (are drivers slowing down? is there congestion building?), road surface conditions (wet, soft, dusty), light conditions (sun glare, dusk, artificial lighting adequacy), and crew state (fatigue, distraction, morale).

Reading conditions is not passive observation. It is active scanning — deliberately looking for what has changed, not just what is there. The experienced TC who notices that the wind has strengthened enough to make a large sign unstable is reading conditions. The TC who walks the site on arrival and spots a pothole that wasn't there last week is reading conditions. The supervisor who sees a crew member struggling with fatigue at the two-hour mark is reading conditions.

In Hollnagel's Resilience Engineering framework, reading conditions corresponds to the "monitor" capacity — the ability to know what to look for and to recognise when something is changing. It is the foundation of adaptive capacity. Without it, adjustment is impossible because you don't know what you're adjusting to.

2. Adjusting Controls

When conditions change, do you actually change what you're doing? Or do you push through because "the TMP says so"? Adjusting controls means having both the authority and the willingness to modify the traffic management setup in response to what you observe.

This is where the framework connects directly to existing standards. The MRWA TMP risk register section 3.2.1.2 states: "Traffic Management Personnel may adjust or change the site where a significant safety hazard is identified with appropriate approval and risk assessment." This sentence, buried in a risk register, is the regulatory basis for dynamic awareness. The framework gives it prominence and process.

Adjusting controls might mean repositioning a TC who can't see oncoming traffic due to sun glare. It might mean reducing the speed zone when drivers are ignoring the posted limit. It might mean adding a spotter when visibility drops. It might mean stopping the job entirely until conditions improve. The adjustment is always proportional to the observed change. In Hollnagel's framework, this corresponds to the "respond" capacity — knowing what to do when something happens.

3. Stop Authority

The willingness to call a halt. Not after something goes wrong — when you feel something is off. This is where culture matters more than paperwork. If a TC feels they cannot call a stop without being reprimanded, the entire practice collapses. Stop authority is not a power — it is a permission that must be granted by the organisation, reinforced by supervisors, and exercised by every crew member without consequence.

The DuPont safety culture model operates on a principle that any worker can stop any job at any time without question. This is not a sign of weakness or disruption — it is a sign of a mature safety culture. In

Conklin's HOP framework, stop authority is an expression of the principle that error is normal and systems must be designed to tolerate it. When a TC calls a stop, they are not causing disruption — they are exercising the adaptive capacity that prevents a near-miss from becoming an incident.

The Thread: Communication

Communication is not a fourth pillar or a fifth element. It is the thread that runs through all four components of the framework. The three pillars are verified through communication: pre-start briefings, equipment checks, plan reviews. Dynamic awareness is practised through communication: calling out what you see, radioing changes, speaking up about concerns, and — critically — calling a stop when something feels wrong.

Communication failures appear in banner alerts not as standalone causes but as contributing factors that amplify other failures. In EQ#56820, the prestart form was a communication failure — a form completed without genuine engagement. In EQ#57801, the chain recovery was a communication failure — nobody flagged that proper recovery equipment was not available. In each case, communication was the mechanism that could have bridged the gap between what was planned and what was actually happening.

For this reason, communication is not positioned as a separate pillar. Making it one would risk reducing it to a checkbox — "radio check, tick" — rather than recognising it as the continuous process that enables every other element of the framework. The DuPont FLM course teaches the "5-Minute Conversation" — short, focused safety conversations that are not checklists but genuine observations: "I noticed you repositioned the taper — what did you see?" This is communication as a skill to teach and practise, not a box to tick.

5. The Cultural Reproduction Problem

The hardest question facing any safety framework is not "what should good practice look like?" but "how do you change practice when the people transmitting the culture are the problem?" This section addresses that question directly, because no framework — Safety I, Safety II, or anything else — can succeed if the people leading crews are modelling the wrong behaviour.

5.1 The Apprenticeship of Bad Practice

When a new Traffic Controller arrives on their first job, they are a blank slate. They want to do a good job, but they don't yet know what good looks like. What they learn is whatever they see. If the crew leader does the prestart properly, walks the site, discusses hazards, and monitors conditions throughout the shift, the new TC learns that this is how professionals work. If the crew leader mumbles through the

prestart, takes a phone call halfway through, doesn't walk the site, and spends the shift on their phone, the new TC learns something very different: this is how professionals work.

The problem is not that the new TC is lazy or disengaged. The problem is that they are doing exactly what humans do: learning from observation. They assume the person leading them is the standard. And why wouldn't they? The crew leader has more experience, more tickets, and more authority. If this is how they do it, this must be how it's done.

Over a year, a single disengaged crew leader might influence 20-30 different TCs who rotate through their crew. Each of those takes the model to their next crew. Within three years, one crew leader's minimum-standard approach has infected an entire labour pool. This is not a training problem — it is a cultural reproduction problem, and it is exponential.

5.2 The Normalisation of Deviance

The sociologist Diane Vaughan coined the term "normalisation of deviance" in her study of the Challenger space shuttle disaster. She did not mean people were deliberately deviating from standards. She meant that gradually, incrementally, what was once recognised as substandard becomes accepted as normal, because everyone sees everyone else doing it and nobody says anything. The standard doesn't drop in one step. It erodes.

In TC work, the erosion looks like this: the original standard is a full prestart with a site walk, hazard discussion, and continuous monitoring. The first erosion shortens the prestart and replaces the site walk with a drive-past. The second erosion makes the prestart a formality and drops the hazard discussion. The third erosion turns the prestart into a checkbox and replaces monitoring with "text me if something happens." The new TC who arrives at the third erosion thinks this is the standard. They have never seen the original.

5.3 Why Compliance Systems Make It Worse

Here is the most perverse aspect of the current system: compliance mechanisms actually accelerate the normalisation of deviance. When a crew leader ticks the prestart box without genuinely conducting the prestart, two things happen simultaneously. First, the record says it was done — the audit trail is clean, the company sees compliance, no flags are raised. Second, the TC sees it being faked — and they learn that the paperwork matters more than the practice. They learn that safety is performative. They learn that the system can be gamed, and that everyone knows it, and that nobody cares as long as the form is signed.

This is worse than having no system at all. A system that can be gamed teaches people to game systems. It trains cynicism. And cynical TCs do not practise Dynamic Awareness because they have already concluded that safety is theatre. More checklists, more audits, and more forms will not fix this — the disengaged crew leader will tick those boxes too. Compliance is the language of minimums, and minimums are what got us here.

5.4 What the Framework Does Differently

The 3 Pillars, 1 Practice framework addresses cultural reproduction in three ways that the current compliance system does not:

- **The 3 Pillars create a visible floor:** The three pillars are verification questions, not checkbox declarations. "Can you demonstrate the skills for this specific site?" cannot be faked the way "is the crew competent?" can be ticked. Verification requires demonstration, not declaration.
- **Dynamic Awareness creates a visible ceiling:** Dynamic Awareness gives the new TC something to look up to. Right now, the new TC has no model of what good looks like beyond the checklist. Dynamic Awareness says: "The standard isn't 'set up and stand there.' The standard is 'set up, then keep reading the job.' Here's what that looks like." Even if the person next to them isn't hitting that standard, the new TC at least has a conceptual target.
- **The organisational layer creates accountability for modelling:** The framework explicitly holds organisations accountable for ensuring that crew leaders can model the ceiling, not just meet the floor. If a crew leader cannot demonstrate Dynamic Awareness, the organisation has failed — not the individual TC who copies them. This is a fundamental shift from the current system, which holds individuals accountable for compliance but never holds the organisation accountable for the conditions that produce non-compliance.

6. Monitoring What Matters: From Compliance Audits to Behavioural Visibility

If the framework is going to work, there must be a way to know whether it is being practised. But here is the central challenge: bad behaviour is easy to monitor, good behaviour is hard. A cone out of position can be photographed. A sign facing the wrong way can be ticked as a non-conformance. But how do you monitor vigilance? How do you audit situational awareness? How do you record the fact that a TC was actively scanning the approach every 30 seconds?

6.1 Why Monitoring Good Behaviour Is Different

The DuPont behavioural observation programs that were widely adopted in the 1990s and 2000s often failed because they turned behavioural safety into a surveillance exercise. Workers performed for the observer rather than working naturally. The observation changed the behaviour being observed. This is a well-documented phenomenon in safety research, and it explains why checklist-based behavioural observation programs produce data but not insight.

Safety II offers a different approach. Instead of monitoring for compliance, monitor for capacity. Instead of auditing what went wrong, study what goes right. Instead of watching workers, listen to them. The practical techniques below are drawn from Hollnagel, Conklin, and Dekker, adapted for the specific conditions of temporary traffic management.

6.2 Practical Techniques for TC Operations

The "What Did You See?" Conversation

Not an audit. Not a checklist. A genuine question, asked with curiosity, not judgment: "I noticed you moved the stop/slow downstream — what did you see?" "You called for a hold before that truck came through — what told you to do that?" The answers reveal the Dynamic Awareness process. The act of asking validates it. Over time, this creates a culture where TCs expect to be asked about their thinking, not just their compliance. This is not a management function — it is a crew function. When peers ask each other, it builds interdependence. When only supervisors ask, it builds dependence.

Positive Observations

Most TC companies have Non-Conformance Report systems. Almost none have a system for recording positive observations. Build one. "Observed: TC repositioned taper approach due to wind gusts exceeding 40km/h. Adjusted cone spacing to next standard up. Maintained traffic flow. Good dynamic assessment." This does two things: it creates a record of good practice (evidence for the organisation), and it signals to the crew that someone notices when they do the right thing, not just when they do the wrong thing.

The Close Call Log

"Near-miss reporting" carries negative connotation — something almost went wrong. Reframe it as a Close Call Log — something went right because someone noticed. "Close call: Wind started blowing cones into live lane. TC on east approach noticed and weighted cones before any entered traffic. No incident." The framing is critical: the good behaviour prevented the incident. The close call log documents the presence of safety capacity, not the absence of an incident.

Crew Self-Assessment

After each job, the crew rates themselves on the 3+1. Not a supervisor rating — self-rating. Competent Crew: 4 out of 5. Sound Plan: 5 out of 5. Right Equipment: 3 out of 5 — weighted bases were at other depot. Dynamic Awareness: 4 out of 5 — caught the wind shift early but missed the sun glare until it was already a problem. This is self-monitoring, which is the Independent stage on the Bradley Curve. When crews start sharing their self-assessments with each other and learning across crews, that is Interdependent.

Peer Observation

DuPont-style behavioural observation has a supervisor watching workers. Safety II flips this: TCs observe each other. Not to catch mistakes — to learn from each other's practice. "I watched how you managed that queue — I'd have done it differently. Can I ask why you chose that position?" This normalises conversation about decision-making. It makes Dynamic Awareness discussable. The key difference: peer observation is horizontal (crew to crew), not vertical (management to worker). It builds interdependence. Management observation builds dependence.

6.3 The Technology Opportunity

Current safety technology in the TC industry focuses almost entirely on compliance — is the TMP signed, is the SWMS on site, is the audit complete. No existing system captures evidence of Dynamic Awareness in real time. This represents a significant opportunity. The TC Work Zone Locator app, for example, already has capabilities that could be directed toward making good safety behaviour visible:

App Capability	What It Captures	Good Behaviour Evidence
GPS + speed zone lookahead	TC checks speed zone before arrival	Proactive verification (Pillar 2)
Live weather feed	Crew notes wind change, adjusts setup	Dynamic Awareness — reading conditions
Banner alerts near SLK	Crew reviews nearby alerts during prestart	Dynamic Awareness — learning from others
AfterCare lookahead	Crew positions with awareness of downstream	Dynamic Awareness — anticipating
Close call logging (new)	TC logs a close call in real time	Evidence of active monitoring and response

Nobody in the TC industry is capturing this data. The first system that does will have a significant advantage — not just as a compliance tool, but as a safety culture tool that makes good practice visible and measurable for the first time.

7. Where This Lives in Existing Standards

The framework does not introduce new obligations. Every element maps to existing requirements in Australian and Western Australian standards. What the framework provides is structure — a way to organise existing obligations so that the gap between planning and practice becomes visible.

Framework Element	Existing Standard Reference	What the Framework Adds
Pillar 1: Competent Crew	AGTTM Part 7 — TC competency requirements MRWA CoP Section 8 — Accreditation WHS Act S27 — Officer due diligence	Distinguishes accreditation from competence. Makes context-specific experience visible, not just card status.
Pillar 2: Sound Plan	MRWA CoP Section 4 — TMP requirements AGTTM Part 3 — TGS design TMP 0922 Section 3.2.1 — Plan adequacy	Requires the plan to be verified as current and site-specific, not just present. Adds crew understanding as a criterion.
Pillar 3: Right Equipment	MRWA CoP Section 6.16 — Speed feedback signs MRWA CoP Section 6.3 — PPE MRWA CoP Section 6.14 — Implementation and maintenance	Requires serviceability and sufficiency, not just presence. Adds contingency equipment.
Practice: Dynamic Awareness	TMP 0922 Section 3.2.1.2 — Site adjustment authority TMP 0922 Section 3.2.2 — Environmental conditions WHS Act S19 — Primary duty of care	Names and structures what is currently one sentence in a risk register. Provides the three-component model (read, adjust, stop). Grounds it in Safety II.
Thread: Communication	MRWA CoP Section 6.14 — TTM operation AGTTM Part 7 — Communication protocols WHS Act S19 — Duty to consult	Positions communication as the enabler of all other elements, not a standalone compliance item.

The key mapping is between Dynamic Awareness and TMP Section 3.2.1.2, which states that Traffic Management Personnel may adjust or change the site where a significant safety hazard is identified. This single sentence is the regulatory authorisation for dynamic awareness. The framework transforms it from a buried provision into a visible, teachable, measurable practice.

8. The Maturity Model

Not all organisations practise the framework equally. The maturity model describes four levels of adoption, drawing on the Bradley Curve used in industrial safety culture assessment. It provides organisations with a way to assess where they currently sit and what the next step looks like.

Level	Name	Pillars Present	Practice Present	What This Looks
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				Like
1	Compliance Only	2 and 3 (signs and plan)	No	The minimum. Signs are loaded, the TMP is on the dashboard. Crew competency is assumed based on accreditation. No process for reassessment during the job.
2	Competent Individuals	1, 2, and 3 (crew, plan, equipment)	Informal only	Qualified crew doing the right thing individually. Experienced TCs may practise dynamic awareness personally, but it's not shared or supported by the organisation.
3	Coordinated Team	1, 2, and 3 (all verified)	Partially (no stop authority)	Crew communicates and coordinates effectively. Conditions are discussed. But stop authority is unclear or conditional — TCs hesitate to call a halt.
4	Adaptive Crew	1, 2, and 3 (all verified)	Yes (read, adjust, stop)	The full framework. The team reads conditions, adjusts controls when needed, and has clear stop authority. This is not dependent on individual experience — it is embedded in the crew's culture.

An organisation at Level 2 can pass an audit. An organisation at Level 4 is where nobody gets hurt. The difference is not in the paperwork — it is in the practice.

Most traffic management companies in Western Australia currently operate between Level 1 and Level 2. They have the signs and the plan. Some have competent crews. Very few have systematically addressed dynamic awareness as a safety practice, and even fewer have explicitly granted stop authority to their Traffic Controllers.

The maturity model also maps to the Bradley Curve stages. Level 1 corresponds to the Dependent stage — people follow rules because they are told to. Level 2 corresponds to early Independent — individuals take personal responsibility but don't share it. Level 3 is late Independent — the team coordinates but lacks the cultural permission to challenge. Level 4 is Interdependent — the team monitors itself, adapts collectively, and has full stop authority. Dynamic Awareness is the behavioural expression of the Interdependent stage. It cannot be achieved through compliance alone; it requires a cultural shift that the organisation must lead.

9. What This Means for Organisations

9.1 The Two Layers

The framework operates at two layers. The crew layer is what the Traffic Controller can control and practice: verifying the three pillars at the pre-start, practising dynamic awareness throughout the shift, and communicating continuously. The organisational layer is what the company must provide: the systems, resourcing, and culture that make the crew layer possible.

The banner alerts demonstrate what happens when the organisational layer fails. In EQ#56820, the prestart form was reused for a week because the company's system allowed it. In EQ#57801, proper recovery equipment was not available because the company did not equip the crew for contingencies. In each case, the crew could not practise what the framework requires because the organisation had not provided the conditions for it.

Layer	Responsibility	Examples
Crew	What the TC can control and practice	Verifying pillars at pre-start Reading conditions during the job Adjusting controls when conditions change Calling a stop when something feels wrong Communicating observations to the team
Organisational	What the company must provide	Systems that ensure plans are current and task-specific

		Resourcing that puts competent supervision on every job Culture that grants stop authority without consequence Learning loops that feed incidents back into practice Equipment that goes beyond the minimum schedule Crew leaders who model Dynamic Awareness, not just compliance
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9.2 Breaking the Chain: Practical Interventions

The only way to stop cultural reproduction of bad practice is to interrupt the apprenticeship. The following interventions are designed to be practical, low-cost, and implementable without new systems or new paperwork:

Separate "Buddy" from "Model"

Currently, new TCs are assigned to whichever crew has a spare spot. The person they learn from is random. Instead, designate mentors — experienced TCs specifically selected and trained to model Dynamic Awareness. Mentor criteria should not be "has been doing this for five years" but "can describe their decision-making process in real time" and "has demonstrated Dynamic Awareness under observation." New TCs should spend their first 20 jobs with mentors only. This breaks the transmission chain because the first 20 jobs establish the normal.

Make the Invisible Visible

The new TC copies the crew leader because they assume the crew leader is the standard. Make the actual standard visible through briefing cards — simple, pocket-sized cards that describe what Dynamic Awareness looks like in practice: "Scan the approach every 30 seconds. Check the sky for weather changes. Listen for speed changes. Reposition if conditions shift." These are not checklists — they are descriptions of behaviour, not lists of tasks. The difference matters. A checklist gets ticked. A behaviour description gets practised.

Raise the Crew Leader Standard

Not everyone with a traffic management ticket should be a crew leader. Currently, crew leader selection is based on having the right tickets, being available, and not being the newest person. It should be based on whether they can model Dynamic Awareness, can explain their decision-making to a new TC, can run a prestart that is a conversation rather than a monologue, and can handle a situation that is not in the TMP. If the answer to "will new TCs be better after working with you or will they learn your shortcuts?" is the latter, that person should not be crew leading.

The "What Did You See?" Protocol

Every crew, every job, every handover: the outgoing crew leader asks the incoming crew leader "What did you see today?" Not "how was it?" — that gets "fine." "What did you see?" requires a specific answer about observations, adjustments, and changes. The new TC hears this exchange and learns that talking about what you noticed is normal. It takes 30 seconds. It is not a meeting. It is not paperwork. It is a conversation that models the behaviour you want.

9.3 Implementation Without Burden

The framework is designed to be adopted without adding compliance burden. It does not require new forms, new checklists, or new reporting obligations. It requires a shift in how existing processes are used:

- **Pre-start briefs:** The pre-start brief already happens. The framework asks that it covers all three pillars explicitly, not just the plan.
- **Site adjustments:** Site adjustments are already permitted under TMP 3.2.1.2. The framework asks that they are made proactively, not only after an incident.
- **Stop authority:** Stop authority is already implicit in the WHS duty of care. The framework asks that it is made explicit — every crew member knows they can call a stop, and every supervisor backs them up.
- **Incident learning:** Banner alerts are already issued by MRWA. The framework asks that they are used as learning tools, not just filed as compliance documents.

The shift is cultural, not procedural. The three pillars are already being checked in most organisations, though often implicitly. The practice of dynamic awareness is already being performed by experienced TCs, though often unconsciously. The framework makes both visible, teachable, and measurable — without requiring a new system to do it.

9.4 The Business Case

For organisations, the framework offers a clear business case beyond the moral imperative of keeping workers safe:

- **Reduced incidents:** Every Lost Time Injury costs the organisation in workers' compensation, lost productivity, investigation time, and replacement labour. The banner alerts document LTIs that occurred on jobs where the preconditions were met — meaning these were preventable with dynamic awareness, not with more paperwork.
- **Regulatory positioning:** Organisations that can demonstrate a mature safety culture — including the practice of dynamic awareness and explicit stop authority — are better positioned for MRWA pre-qualification, contract evaluation, and insurance assessment.
- **Crew retention:** TCs who feel empowered to speak up and adjust controls are more engaged, more likely to stay with the organisation, and more likely to develop into supervisors. The framework supports retention by valuing the judgement that experienced TCs already exercise.

- **Due diligence:** The WHS Act 2020 Section 27 requires officers to exercise due diligence, including acquiring and keeping up-to-date knowledge of hazards and risks. Dynamic awareness is the operational expression of that duty — it is how an organisation demonstrates that it is managing risks in real time, not just on paper.

10. The Evidence: Banner Alerts

Mapped to the Framework

The evidence base for this framework has been expanded from 7 to 64 MRWA Banner Alerts. Each alert has been classified as either directly or indirectly related to traffic control (TC) for filtering purposes. This classification allows the framework to distinguish between incidents that occur within the traffic control system itself and those that occur in the broader work environment but still affect TC workers.

10.1 Overview of the Evidence Base

The 64 banner alerts span the period from November 2023 to April 2026 and include the full range of incident types reported through the MRWA Banner Alert system. The following key statistics describe the dataset:

- **Total alerts:** 64 (45 Red, 19 Grey, 0 Amber)
- **Directly related to TC:** 25 (39.1%)
- **Indirectly related to TC:** 39 (60.9%)
- **LTI count in direct alerts:** 3

Note: The direct/indirect classification is not a judgement on severity. Indirect alerts can and do result in serious injuries. The classification is a filtering tool designed to identify which alerts speak most directly to the traffic control system itself, and therefore which alerts are most relevant to validating the 3 Pillars, 1 Practice framework.

10.2 Direct TC Alert Subcategories

The 25 directly TC-related alerts fall into five subcategories, each representing a distinct type of interaction with the traffic control system:

MoP Breach (10 incidents, 40%)

The largest subcategory of directly TC-related incidents involves Members of Public breaching or interacting dangerously with traffic control setups. These incidents are significant because they represent the category of risk that compliance systems are least equipped to manage. A TMP cannot

control driver behaviour; a TGS cannot prevent a motorist from driving through a Vehicle Control Point or veering into oncoming traffic. The 10 incidents range from a learner driver overtaking and causing a maintenance truck to veer off the road (EQ#42098), to a heavy vehicle losing load and damaging a TC light vehicle (EQ#38338), to a motorist driving through a Vehicle Control Point and colliding with a TC and Fire Services (EQ#37689). Several involve vehicles veering into work zones or making unexpected manoeuvres within TC setups. These incidents validate the framework's emphasis on Dynamic Awareness: crews must continuously read traffic behaviour and adjust controls in real time, because the plan assumes public compliance but reality frequently proves otherwise.

TM Procedure Breach (8 incidents, 32%)

The second largest subcategory involves breaches of traffic management procedures, most commonly workers found within 1.2 metres of live traffic without appropriate controls. The 1.2m rule appears in 5 of the 8 TM breaches, suggesting this is a critical risk area where the gap between the rule and the reality of getting work done near live traffic is widest. Critically, several of these breaches occurred not because workers were reckless, but because of systemic pressures: time pressure to complete core testing before the rostered swing ended (EQ#39231, EQ#39234), inadequate supervision when project engineers departed site early (EQ#39231), and failure to follow the communications of the Traffic Management Coordinator (EQ#39486). Other TM breaches include incorrect Traffic Guidance Scheme implementation (EQ#43968), a Traffic Controller on foot within 1.2m in a 60km speed zone (EQ#35211), and a breach of the Mobile Plan Onboarding Procedure (EQ#38716). These are not individual failures but system failures that individual compliance alone cannot prevent.

TC Injured (4 incidents, 16%)

Four incidents involve Traffic Controllers physically injured while performing their duties. Three of these resulted in Lost Time Injuries, making this the subcategory with the highest LTI rate. The incidents include a TC found lying face down with concussion (EQ#37292), a TC experiencing back pain performing manual labour tasks (EQ#40656), a TC working excessive hours without sufficient breaks between shifts (EQ#41942), and a TC sustaining a shoulder injury slipping on a roadside batter (EQ#57872). The pattern is telling: TCs are being injured by the physical demands and environmental conditions of the role, not by the traffic they are controlling. The TC found face down with concussion is particularly concerning because the alert does not specify how the injury occurred, which itself indicates the isolated nature of TC work. Dynamic Awareness, in this context, means crew members watching out for each other, monitoring fatigue, and recognising when a colleague is at risk.

TC Equipment (2 incidents, 8%)

Two incidents involve TC-specific equipment creating hazards. In EQ#37443, a follow-me vehicle conducted a U-turn in front of a road train, creating a high-potential collision scenario. In EQ#37835, a TC sustained a lower back strain accessing a Truck Mounted Attenuator (TMA), resulting in an LTI. These incidents highlight that the equipment designed to protect traffic controllers can itself become a source of risk, particularly during mounting, dismounting, and repositioning operations. The follow-me vehicle

incident scored 4 on the framework relevance scale (conditions changed, crew competency gaps, plan adequacy failures, and equipment issues all flagged), indicating multiple pillar failures converging. The TMA access injury demonstrates that ergonomic hazards in TC-specific equipment are an under-recognised risk category.

TM Setup (1 incident, 4%)

One incident involves a worker entering a live lane to place traffic signage (EQ#48096). While only a single incident in this dataset, it highlights one of the most hazardous moments in traffic management: the setup and pack-down phases, when the full TM layout is not yet in place or has been partially removed. During these phases, workers are most exposed to live traffic because the protective framework of signs, cones, and speed reductions is incomplete. The framework relevance score of 1 (only the paperwork-vs-reality gap flagged) suggests this is an area where the plan assumes safe setup procedures but reality creates pressure to take shortcuts. This single incident should not be dismissed as an outlier; setup and pack-down are universally recognised in the industry as the highest-risk phases of any traffic management job.

Subcategory	Count	Description
mop_breach	10	Member of Public breaching or interacting with TC setup/work zone
tm_breach	8	Traffic management breach (1.2m rule, wrong TGS, TM procedure violation)
tc_injured	4	Traffic Controller injured while performing TC duties
tc_equipment	2	TC equipment involved (TMA, follow-me vehicle)
tm_setup	1	TM setup/removal activity (placing signage, entering live lane)

10.3 Indirect TC Alert Subcategories

The 39 indirectly TC-related alerts fall into eleven subcategories, representing hazards that affect TC workers but are not inherent to the traffic control system itself:

Journey Management (7 incidents)

The largest indirect subcategory involves journey management failures, including contractors failing to arrive at site prompting search and rescue, vehicles hitting livestock, and drivers falling asleep at the wheel. These incidents highlight that the risks to TC workers extend well beyond the work zone itself.

Remote and regional Western Australian roads present significant hazards for the drive to and from sites, often involving long distances, wildlife, and fatigue. The journey management failures are relevant to the framework because they affect Pillar 1 (Competent Crew): a worker who has driven four hours to reach site may not be in a fit state to manage traffic safely, regardless of their accreditation.

Equipment and Tool Incidents (6 incidents)

Six incidents involve equipment and tool failures or misuse, including workers injured by angle grinders, quick-cut saws, and other equipment. While not directly related to traffic control operations, these incidents are relevant because TC crews regularly handle signs, cones, TMA units, and other equipment that can cause injury. The framework's Pillar 3 (Right Equipment) extends beyond traffic control devices to include all equipment used on the job, and these incidents demonstrate that equipment hazards are a persistent risk across all work types on Main Roads projects.

Slip and Trip Incidents (5 incidents)

Five incidents involve workers slipping or tripping on surfaces, equipment, or environmental hazards. TC work is inherently outdoor and ground-level work, often performed on road shoulders, batters, and unsealed surfaces. The TC who slipped on a roadside batter (EQ#57872) is classified as a direct TC injury, but the pattern extends to other workers on road projects. Slip and trip hazards change with weather, time of day, and surface conditions — exactly the kind of dynamic risk that Dynamic Awareness is designed to address.

Manual Handling (5 incidents)

Five incidents involve manual handling injuries, including strains from lifting, carrying, and positioning heavy objects. TC work is physically demanding: a single 900mm regulatory sign weighs approximately 7 kg, and a typical shift may involve setting up and removing dozens of signs and dozens of cones. The TC who experienced back pain performing manual labour (EQ#40656) is classified as a direct TC injury, but the pattern is broader. These incidents connect to Pillar 3 (Right Equipment) when mechanical aids or alternative methods could reduce manual handling risk.

Fitness for Duty (4 incidents)

Four incidents involve fitness for duty failures, including positive BAC results, a driver who fell asleep at the wheel after taking illicit substances and getting only four hours of sleep (EQ#40634), and workers not declaring medications that could impair their ability to work safely. These incidents directly challenge Pillar 1 (Competent Crew): accreditation without fitness for duty is a broken pillar. The supervisor in EQ#40634 was aware of the worker's fatigue issue in the morning but did not put suitable controls in place, demonstrating that fitness for duty is a shared responsibility between the worker and the organisation.

Mechanical Failures (3 incidents)

Three incidents involve mechanical failures on vehicles and mobile plant, including a leaf spring dislodging from a trailer and injuring a member of public (EQ#39559) and other vehicle component failures. While not directly TC-related, these incidents are relevant because TC operations depend on vehicles (TMA trucks, shadow vehicles, follow-me vehicles) that are subject to the same mechanical risks. The corrective action from EQ#39559 — to include leaf spring checks as part of daily pre-starts — illustrates how mechanical failures can be addressed through the pre-start verification that Pillar 3 (Right Equipment) requires.

Road Furniture (3 incidents)

Three incidents involve road furniture maintenance and installation, including a worker injured whilst repairing wire ropes resulting in LTI (EQ#40534). Road furniture work is adjacent to TC operations: wire rope barriers, guide posts, and signage are part of the roadside environment that TCs work within. The corrective actions from these incidents — ensuring correct equipment availability, sufficient crew resources, and following SWMS — mirror the framework's three pillars exactly.

Construction Activities (2 incidents)

Two incidents involve construction-phase activities on road projects. These are indirectly relevant to TC because construction activities occur within or adjacent to TC setups, and the safety of construction workers depends partly on the quality of the traffic management protecting them. The boundary between "TC" and "construction" incidents is often blurry: a worker within 1.2m of live traffic because of inadequate TM is both a TM breach and a construction risk.

Vehicle Incidents (2 incidents)

Two incidents involve vehicle collisions or loss of control not directly within a TC zone. These are indirectly relevant because they demonstrate the general road environment risk that TC operations must manage, and because TC vehicles are exposed to the same hazards when travelling to and from sites.

Utility Strike (1 incident)

One incident involves an excavator operator striking an unidentified overhead power line (EQ#40675). While not a TC incident per se, it is relevant because TC operations must account for overhead and underground utilities when positioning signs and TMA units. The contributing factors — inadequate BYDA plans, new operator without adequate supervision, and unmarked infrastructure — parallel the TC-specific risks of working in environments where not all hazards are identified in the plan.

Environmental (1 incident)

One incident involves environmental factors creating workplace hazards. While a single data point, it reinforces that the outdoor, uncontrolled environment in which TC work takes place is subject to environmental conditions that change continuously — precisely the type of risk that Dynamic Awareness is designed to address.

Subcategory	Count	Description
journey	7	Journey management incident (driving to/from site)
equipment_tool	6	Equipment or tool injury
slip_trip	5	Slips, trips, falls on site
manual_handling	5	Manual handling injury (back strain, lifting, pulling)
fitness_for_duty	4	Fitness for duty (drug/alcohol, medical episode, fatigue)
mechanical	3	Mechanical failure (leaf spring, chain, vehicle component)
road_furniture	3	Road furniture maintenance (guide posts, wire ropes)
construction	2	Construction activity injury
vehicle_incident	2	Vehicle incident not in work zone or TM-related
utility_strike	1	Utility strike (power line, service)
environmental	1	Environmental hazard (thermal stress, weather)

10.4 Detailed Direct TC Alert Register

The following table lists all 25 directly TC-related banner alerts, including their classification, subcategory, description, and LTI status:

EQ#	Date	Banner	Subcategory	Description	LTI	Pillars Flagged	Score
35211	2023-11-08	Grey	Tm Breach	Traffic Controller on foot within 1.2m in a 60km speed zone	No	Plan, Equip, Gap	3
35216	2023-11-13	Grey	Mop Breach	MOP collided with Main Roads Employee after veering into	No	Crew, Plan, Equip, Gap, Change	5

				oncoming lane			
37100	2024-01-20	Red	Tm Breach	Mobile Plant and Traffic Management Procedure Breach	No	Plan, Equip, Gap	3
37292	2024-01-30	Red	Tc Injured	Traffic Controller found lying face down with concussion	Yes	Plan, Equip, Gap	3
37443	2024-02-04	Grey	Tc Equipment	Follow me vehicle conducts U-turn in front of road train	No	Crew, Plan, Equip, Change	4
37689	2024-02-08	Red	Mop Breach	MoP driven through Vehicle Control Point colliding with TC and Fire Services	No	Equip	1
38338	2024-02-23	Red	Mop Breach	MoP Heavy vehicle loses load and damages TC light vehicle	No	Plan, Equip	2
37835	2024-02-24	Red	Tc Equipment	Lower back strain accessing TMA resulting in LTI	Yes	Equip, Gap, Change	3
38716	2024-03-08	Red	Tm Breach	Breach of Mobile Plan Onboarding Procedure	No	Plan, Equip, Gap	3
38753	2024-03-11	Red	Mop Breach	Member of Public Rollover within Works Area	No	Plan, Equip	2
38369	2024-03-22	Red	Mop Breach	Worker struck by Member of Public	No	Equip	1
39231	2024-03-28	Grey	Tm Breach	Traffic Management Breach - Lab (core) Tester working within 1.2m of live traffi	No	Crew, Plan, Equip, Gap	4
39234	2024-03-28	Red	Tm Breach	Traffic Management Breach - Lab (core) Tester working within 1.2m of live	No	Plan, Equip	2
39486	2024-04-09	Grey	Tm Breach	Traffic Controller and work vehicle within 1.2 of live	No	Crew, Plan, Equip, Gap, Change	5

				traffic lane			
40545	2024-05-21	Red	Tm Breach	Core tester observed working within 1.2m of live traffic travelling at 110km hr	No	Plan	1
40615	2024-05-24	Red	Mop Breach	MOP collides with Contractor Vehicle turning into Site Office	No	Equip	1
40656	2024-05-30	Red	Tc Injured	Traffic Controller experiences back pain whilst performing manual labour tasks	No	Equip, Gap	2
41100	2024-06-11	Red	Mop Breach	OSOM breaches traffic management	No	Plan, Equip	2
41942	2024-06-26	Red	Tc Injured	Traffic Controller working excessive hours without sufficient breaks between wor	No	Crew, Plan, Gap	3
42098	2024-07-02	Red	Mop Breach	Learner Driver overtaking causes maintenance truck to veer from road	No	Crew, Equip, Change	3
43968	2024-09-05	Grey	Tm Breach	Incorrect Traffic Guidance Scheme	No	Plan, Gap	2
47857	2025-02-02	Red	Mop Breach	Heavy Vehicle veers into	No	Equip, Change	2
48009	2025-02-04	Red	Mop Breach	Near miss from light vehicle overtaking 3 stationary trucks and running a Red Li	No	Equip, Gap	2
48096	2025-02-12	Red	Tm Setup	Worker enters live to place traffic signage	No	Gap	1
57872	2026-04-11	Red	Tc Injured	Traffic Controller sustains shoulder injury slipping on roadside batter	Yes	Plan, Equip	2

Pillar frequency across 25 direct alerts: Equipment Issues flagged in 21 incidents (72%), Plan Adequacy Failures in 16 (52%), Paperwork-vs-Reality Gap in 13 (44%), Crew Competency Gaps in 6 (28%), and

Conditions Changed Mid-Job in 6 (20%). The dominance of equipment and plan failures confirms that the three pillars are being checked superficially rather than substantively. The relatively low rate of conditions-changed-mid-job flags (20%) suggests that most TC-direct incidents occur during stable conditions — the plan was wrong from the start, or the equipment was inadequate from the outset, rather than conditions shifting during the job.

EQ#	Date	Banner	Subcategory	Description	LTI
EQ#35211	2023-11-08	Grey	tm_breach	TC on foot within 1.2m in a 60km speed zone	No
EQ#35216	2023-11-13	Grey	mop_breach	MoP collided with MR Employee after veering into oncoming lane	No
EQ#37100	2024-01-20	Red	tm_breach	Mobile Plant and Traffic Management Procedure Breach	No
EQ#37292	2024-01-30	Red	tc_injured	TC found lying face down with concussion	Yes
EQ#37443	2024-02-04	Grey	tc_equipment	Follow me vehicle conducts U-turn in front of road train	No
EQ#37689	2024-02-08	Red	mop_breach	MoP driven through Vehicle Control Point colliding with TC and Fire Services	No
EQ#37835	2024-02-24	Red	tc_equipment	Lower back strain accessing TMA resulting in LTI	Yes
EQ#38338	2024-02-23	Red	mop_breach	MoP Heavy vehicle loses load and	No

				damages TC light vehicle	
EQ#38369	2024-03-22	Red	mop_breach	Worker struck by Member of Public	No
EQ#38716	2024-03-08	Red	tm_breach	Breach of Mobile Plan Onboarding Procedure	No
EQ#38753	2024-03-11	Red	mop_breach	Member of Public Rollover within Works Area	No
EQ#39231	2024-03-28	Grey	tm_breach	TM Breach - Lab (core) Tester working within 1.2m of live traffic	No
EQ#39234	2024-03-28	Red	tm_breach	TM Breach - Lab (core) Tester working within 1.2m of live traffic	No
EQ#39486	2024-04-09	Grey	tm_breach	TC and work vehicle within 1.2m of live traffic lane	No
EQ#40545	2024-05-21	Red	tm_breach	Core tester observed working within 1.2m of live traffic at 110km/hr	No
EQ#40615	2024-05-24	Red	mop_breach	MoP collides with Contractor Vehicle turning into Site Office	No
EQ#40656	2024-05-30	Red	tc_injured	TC experiences back pain whilst performing manual labour	No

				tasks	
EQ#41100	2024-06-11	Red	mop_breach	OSOM breaches traffic management	No
EQ#41942	2024-06-26	Red	tc_injured	TC working excessive hours without sufficient breaks between working shifts	No
EQ#42098	2024-07-02	Red	mop_breach	Learner Driver overtaking causes maintenance truck to veer from road	No
EQ#43968	2024-09-05	Grey	tm_breach	Incorrect Traffic Guidance Scheme	No
EQ#47857	2025-02-02	Red	mop_breach	Heavy Vehicle veers into work area	No
EQ#48009	2025-02-04	Red	mop_breach	Near miss from light vehicle overtaking 3 stationary trucks and running a Red Light	No
EQ#48096	2025-02-12	Red	tm_setup	Worker enters live to place traffic signage	No
EQ#57872	2026-04-11	Red	tc_injured	TC sustains shoulder injury slipping on roadside batter	Yes

10.5 Analysis: What the Evidence Tells Us

MoP Breaches (10 incidents)

The single largest category of TC-direct incidents involves Members of Public interacting with or breaching traffic control setups. This is significant because it is the category of risk that compliance systems are least equipped to manage. A TMP cannot control driver behaviour. A TGS cannot prevent a motorist from driving through a Vehicle Control Point (EQ#37689) or veering into oncoming traffic (EQ#35216). These incidents validate the framework's emphasis on Dynamic Awareness: the crew must continuously read the traffic behaviour around them and adjust controls in real time. The plan assumes compliance from the public; Dynamic Awareness prepares for when that assumption fails.

TM Breaches (8 incidents)

The second largest category involves breaches of traffic management procedures — workers within 1.2m of live traffic, incorrect TGS implementations, and onboarding procedure failures. Critically, several of these breaches occurred not because the workers were reckless, but because of systemic pressures: time pressure to complete core testing before the rostered swing ended (EQ#39231, EQ#39234), inadequate supervision when project engineers departed site early (EQ#39231), and crew not fully aware of hazards or how to implement effective controls as specified in the SWMS (EQ#39486). These are not individual failures — they are system failures that individual compliance cannot prevent. The 1.2m rule appears in 5 of the 8 TM breaches, suggesting this is a critical risk area where the gap between the rule and the reality of getting work done near live traffic is widest.

TC Injuries (4 incidents)

Four incidents involve Traffic Controllers injured while performing their duties — a TC found lying face down with concussion (EQ#37292), a TC experiencing back pain performing manual labour (EQ#40656), a TC working excessive hours without sufficient breaks (EQ#41942), and a TC sustaining a shoulder injury slipping on a roadside batter (EQ#57872). Three of these four resulted in LTI. The pattern here is telling: TCs are being injured by the physical demands and environmental conditions of the role, not by the traffic they are controlling. The TC who was found face down with concussion — the alert does not specify how the injury occurred, which itself is an indicator of the isolated nature of TC work. Dynamic Awareness, in this context, means crew members watching out for each other, monitoring fatigue, and recognising when a colleague is at risk.

TC Equipment and TM Setup (3 incidents)

Two incidents involve TC-specific equipment — a follow-me vehicle conducting a U-turn in front of a road train (EQ#37443) and a TC sustaining a back injury accessing a Truck Mounted Attenuator (EQ#37835). One incident involves a worker entering a live lane to place traffic signage (EQ#48096). These incidents highlight the risks inherent in the setup and pack-down phases of traffic management,

which are often the most hazardous moments of a shift because the full TM layout is not yet in place or has been partially removed.

The Indirect Evidence

The 39 indirectly related alerts provide important context. Journey management incidents (7) represent the largest indirect category, highlighting that the risks to TC workers extend beyond the work zone itself — the drive to and from remote sites is a significant hazard. Fitness for duty incidents (4), including positive BAC results and a driver who fell asleep at the wheel, underscore the importance of Pillar 1 (Competent Crew) being about more than just accreditation — it includes being fit for work. Manual handling and slip/trip incidents (10 combined) point to the physical nature of TC work and the need for environmental awareness that goes beyond traffic management.

Summary

The evidence from 64 banner alerts confirms what the framework predicts: the three pillars (verified at pre-start) are necessary but not sufficient. The 25 directly TC-related alerts show that the most common failure modes — MoP breaches, TM procedure breaches, TC injuries, and equipment/setup incidents — all involve the gap between what the plan assumes and what happens in reality. Dynamic Awareness, the continuous practice of reading conditions and adjusting controls, is the framework's response to this evidence.

11. The Path Forward

This paper is presented for industry consultation. The framework is offered as a structure for a conversation that many in traffic management are already having informally: how do we keep people safe when the plan isn't enough? The 3 Pillars, 1 Practice framework does not claim to have all the answers. It claims to have named the right question.

The framework sits at the intersection of two safety traditions. The three pillars draw on the proven Safety I approach of verification, compliance, and control. The practice of Dynamic Awareness draws on the emerging Safety II paradigm of adaptive capacity, learning from success, and treating workers as the resource rather than the problem. Neither tradition alone is sufficient for the unique conditions of temporary traffic management. Together, they address the structural contradiction at the heart of the industry: we apply Safe System thinking to road users but compliance-based thinking to the workers who implement it.

The practical interventions proposed in this paper — designated mentors, the "What Did You See?" protocol, positive observation systems, crew self-assessment, and close call logging — are designed to be implementable without new systems, new paperwork, or new compliance burden. They require a cultural shift: from monitoring compliance to making good practice visible, from auditing what went

wrong to studying what goes right, and from holding individuals accountable for minimums to holding organisations accountable for creating the conditions where the minimum is not the ceiling.

Feedback is welcomed from Traffic Controllers, supervisors, safety managers, and regulatory bodies. The framework will be refined through industry input before any formal proposal is submitted to Main Roads Western Australia.